

Validation SOP

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0. Introduction: What is Validation?

When a client inspects a vehicle using our app, images are captured and processed by ML models that automatically detect panels and damages. Since model detections require human verification, **validation is the process of reviewing and correcting these detections to ensure the final report is accurate before it reaches the client.**

We have separate environments for EU, US, and AUS clients, and procedures differ by environment.

Each inspection is referred to as a quote. A quote may have multiple stages depending on its flow, which is indicated by its colors (see Table below). In the **VGG Phase**, we review, correct, and annotate the model-detected panels and damages. In the **MC Phase**, we compare the detected damages with the prequote and classify each as New, Existing, or FP.

We have separate environments for EU, US, and AUS clients, and procedures differ by environment.

Note: AUS quotes are limited to the VGG phase; therefore, Sections 4, 5, and 6 are not applicable.

As a validator, you pick quotes from the dashboard and complete the respective phases based on the quote's color:

Quote Color	Required Work
White Quote (Fully Manual Flow)	VGG + MC Page
Red Quote (MC Flow)	MC Page only
Yellow Quote (MV Flow)	VGG only
Pink Quote (LP Validation)	LP Validation only

0.1 Purpose and Scope of SOP

This document is intended for all team members involved in quote validation. It covers the procedures to follow across different validation phases and environments (EU, US, and AUS). Validators must strictly adhere to the rules defined herein.

1. Quote Selection Priority

Always ensure the "Available" toggle is turned ON at the top right of the dashboard to receive quotes, and refresh frequently to avoid missing any quotes (see Fig 1).

Every quote has a deadline based on its **priority level**. Plan your quote selection from the validator dashboard to meet these limits. (see Table 1.1, Table 1.2, and Fig 1).

When multiple quotes have the same priority level, always select the **earliest created quote first** (see Fig 1).

1.1 Priority Levels & Deadlines (EU & US)

Priority	Deadline	Time Logic
P1	Within 10 min of creation	Complete all P1 quotes first. Select quotes from left to right (oldest first).
P2	Within 90 min of creation	After clearing P1, complete P2 quotes. Select from left to right (oldest first).
P3	Same day	Process after all P1s and P2s are cleared, left to right.
P4	Same day	Prioritize after P3, left to right.
P5	Next day	Lowest priority. After all P3/P4s.

1.2 Priority Levels & Deadlines (AUS)

Priority	Deadline	Time Logic
ALL Quotes	Within 30 min of creation	Select quotes from left to right (oldest first).

Sample Quote Selection Process

The dashboard shows a list of quotes ready for validation. The quotes are displayed in a grid with the following details:

- Quote 1:** Quote Id: Q-3u5kes1c0, Mercedes-Benz Sprinter, Status: Segmentation Complete, Number of Images: 21, Dealer: Apollo Logistics Solutions Ltd | 2, 21 Mins ago. **Select Second**
- Quote 2:** Quote Id: Q-rZvaTRXzU, Peugeot Expert, Status: Segmentation Complete, Number of Images: 21, Dealer: SJC Distributions Ltd | 2, 20 Mins ago. **Select Third**
- Quote 3:** Quote Id: Q-B7MJbNDEK, Mercedes Sprinter, Status: Segmentation Complete, Number of Images: 14, Dealer: Everest Logistics | 2, 18 Mins ago. **Select Fourth**
- Quote 4:** Quote Id: Q-653fDWtue1, Opel Vivaro, Status: Segmentation Complete, Number of Images: 21, Dealer: Q SPRINT Transpot and Logistik GmbH | 2, 11 Mins ago. **Select Fifth**
- Quote 5:** Quote Id: Q-10dM6M-gU, Ford Transit, Status: Segmentation Complete, Number of Images: 14, Dealer: FDDS | 2, 6 Mins ago. **Select Sixth**
- Quote 6:** Quote Id: Q-BWEaAy7bzZ, Volkswagen E-Crafter, Status: Segmentation Complete, Number of Images: 6, Dealer: Prepair Check | 1, 3 Mins ago. **Select First**

A blue arrow points to the 'Volkswagen E-Crafter' card with the label 'Priority Number (P1)'.

(Fig 1 Sample Quote Selection Process)

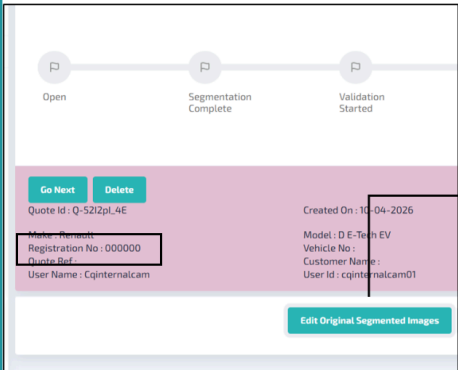
2. LP Validation (Pink Quotes - Fixed Camera Only)

For fixed camera setups, the License Plate (LP) must be verified to ensure the inspection is mapped to the correct vehicle. Review all images and take action based on the situation.

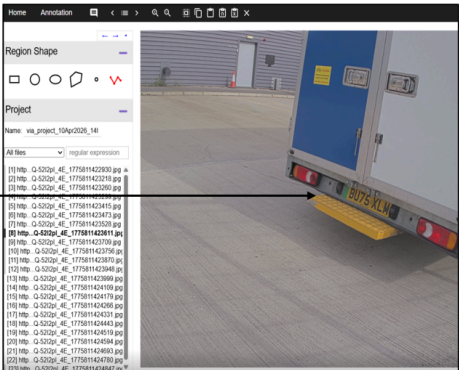
Situation	Details	Action
LP needs correction	Image LP is different from what is detected.	Update LP as per the image, then press "Go Next" (see Fig 2)
Cannot change the LP to match the image	Vehicle belongs to a non-fleet category; cannot be updated in the system.	Delete the inspection
Inspection contains images of two or more different vehicles (different LPs)	Mixed vehicle images; inspection cannot be mapped to a single vehicle.	Delete the inspection
LP is not visible or cannot be identified	Due to poor image quality (e.g., dark or blur images)	Delete the inspection

 Update all deleted quotes with their respective reasons in the [Fixed Camera Quotes Tracking Sheet](#).

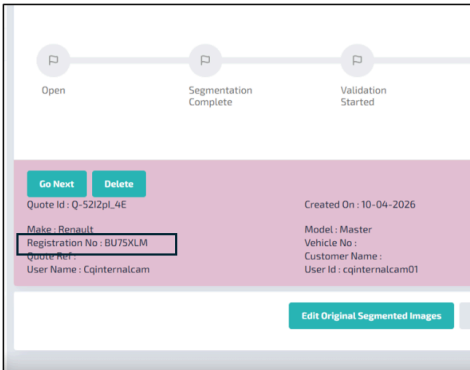
Sample LP Validation Process



Before Correction



Actual LP Verification



After Correction

(Fig 2 Sample LP Validation Process)

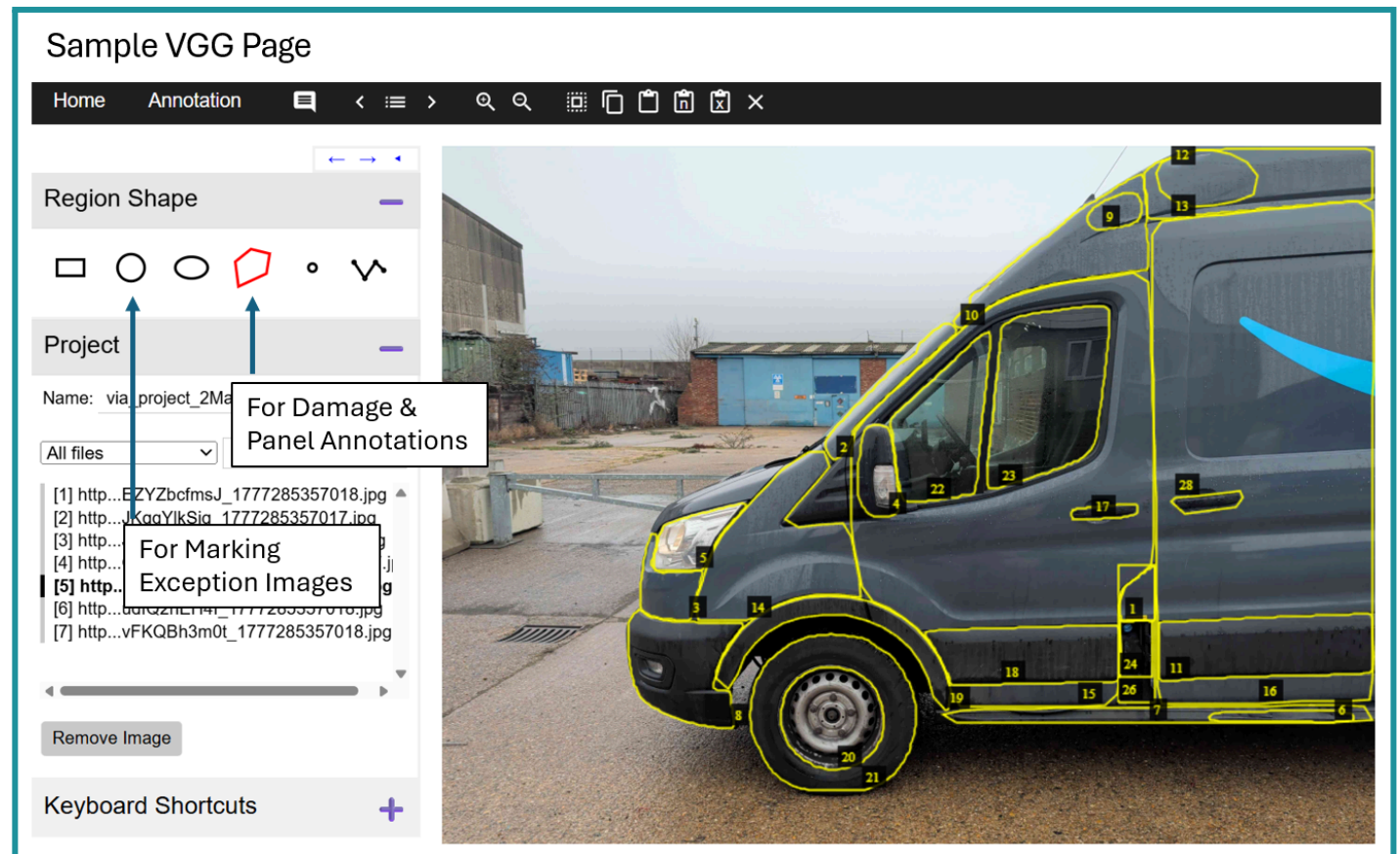
3. Manual Validation (VGG)

Check and correct model-detected damages and panels, and mark any missed damages or panels to ensure report accuracy.

Use the polygon region shape for all Damage and Panel annotations. Use the circular region shape only when marking Exception Images.

3.1 Validation Checklist

Step	What to Do
LP Check (Not Applicable for AUS)	For EU: Verify that the License Plate (LP) is correct; update it if necessary. If the LP cannot be corrected to match the image, report it in the Validation Queries channel. For US: Refer to the US LP Correction Guidelines
Panel Predictions	Check the panel predictions (pre-annotations) and adjust or change them if needed. Manually annotate any panels that the model failed to detect. If panel predictions are inconsistent (e.g., failure to detect major panels, weird annotations, etc.; see Fig 3.1.2), report it for a bodystyle check in the Validation Queries Channel.
Damage Predictions	Identify and manually annotate any missed damages. Check predicted damages; remove FPs and damage transfer FPs. (see Fig 3.1.1)
Potential Panels	Re-review all potential panels (ORVMs, lamps, reflectors, claddings, covers, and trim components) to ensure that no major damage has been overlooked.



(Fig 3.1.1 Sample VGG page)

Sample cases that require a bodystyle check



(Fig 3.1.2 Sample cases that require a bodystyle check)

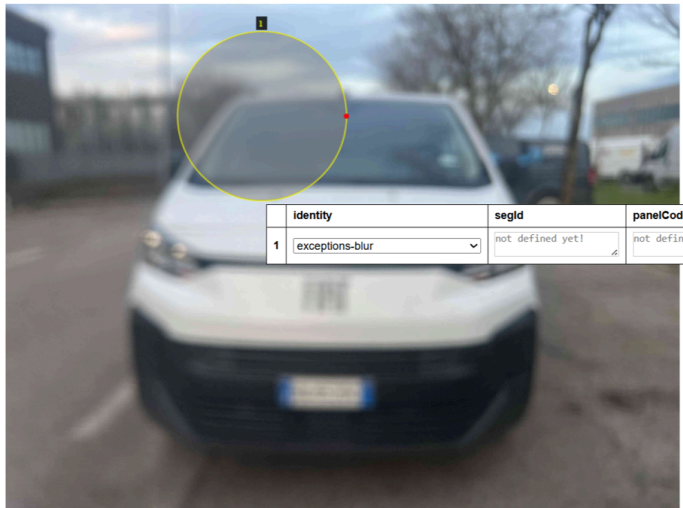
3.2 Image Handling & Exception Categorization

Some images in VGG do not require validation (damage or panel annotations) due to poor quality or exception cases; these must be marked with the respective categories or reported as listed below.

Before marking, you must completely remove any pre-existing annotations (model detections) from the image. (see Fig 3.2.1)

Situation	Action
Blurred image	Mark as exceptions-blur (see Fig 3.2.1)
Dark image	Mark as exceptions-dark (see Fig 3.2.1)
Engine image	Mark as exceptions-engine
Engine number image	Mark as exceptions-enginenummer
Interior image	Mark as exceptions-interiors (see Fig 3.2.2)
Odometer image	Mark as exceptions-odometer
RC image	Mark as exceptions-rc
Spare wheel image	Mark as exceptions-sparewheel
Undercarriage image	Mark as exceptions-undercarriage
VIN image	Mark as exceptions-vin
Screen image (see Fig 3.2.2), Ground image, Nearby vehicle image (nearby vehicle alone) etc	Mark as exceptions-others
Entire quote consists of rotated images	Report to request quote removal in the Validation Queries Channel

Exception Images



exceptions-blur



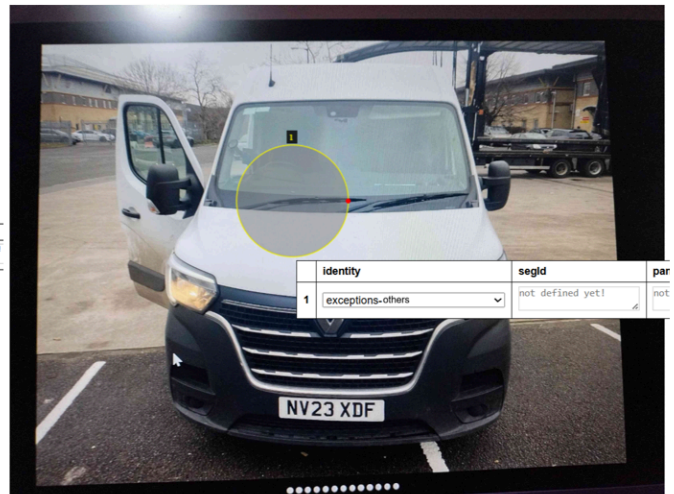
exceptions-dark

(Fig 3.2.1 Exception Images)

Exception Images



exceptions-interiors



exceptions-others (screen image)

(Fig 3.2.2 Exception Images)

3.3 Object-Object & Non-Panel Issues

Situation	Details	Action
[object object] appears in the segId	Happens when you copy-paste a panel annotation for a damage	Remove the segId value and leave it empty . Failing to do so causes comparison failure (see Fig 3.3.1).
Mismatched panelCode after a panel name change	panelCode on the damage annotation still references the old panel name	For non-wheel arch damages, update or remove the panelCode ; for wheel arch damages, the panelCode must be updated.
Damage is outside a panel or outside the vehicle body (see Fig 3.3.2).	Annotation placed incorrectly during detection or validation	Delete the damage.

Damage is located inside a panel but displays "non-panel" in the panelCode (see Fig 3.3.3).	The panel was not predicted, so the model cannot associate the damage to a panel	Annotate the panel. For non-wheel arch damages, update or remove the panelCode; for wheel arch damages, the panelCode must be updated.
An empty list with a null segId and "non panel" panelCode (see Fig 3.3.4)	Might be created with the above mentioned two cases.	Delete the entry.

Object Object Issue

	identity	segId	panelCode
21	detached	[object Object]	leftslidingdoorcladding
21	detached	not defined yet!	leftslidingdoorcladding

(Fig 3.3.1 Object Object Issue)

Outside Non Panel Damage

	identity	segId	panelCode
32	tear	tear_17773507719047774_72_0	non-panel

Before

After

(Fig 3.3.2 Outside Non Panel Damage)

Inside Non Panel Damage

	identity	segId	panelCode
30	d2	d2_1777350771662489088_12.85	non-panel
11	d2	d2_1777350771662489088_12.85	not defined yet!

Before

After

(Fig 3.3.3 Inside Non Panel Damage)

Empty list with a null segld and "non panel" panelCode

Remove Image

23	leftfrontdoorglass	null	leftfrontdoorglass
24		null	non-panel
25	missing	missing_17773507777280 26112_0.0	leftsidefillercladding

Before

Remove Image

22	leftfronttyre	null	leftfronttyre
23	leftfrontdoorglass	null	leftfrontdoorglass
24	missing	missing_17773507777280 26112_0.0	leftsidefillercladding

After

(Fig 3.3.4 Empty list with a null segld and "non panel" panelCode)

Update all poor image quotes with their respective image issues in the [Poor Image Quotes Tracking Sheet](#).

For any other issues or queries, report in the **Validation Queries Channel**.

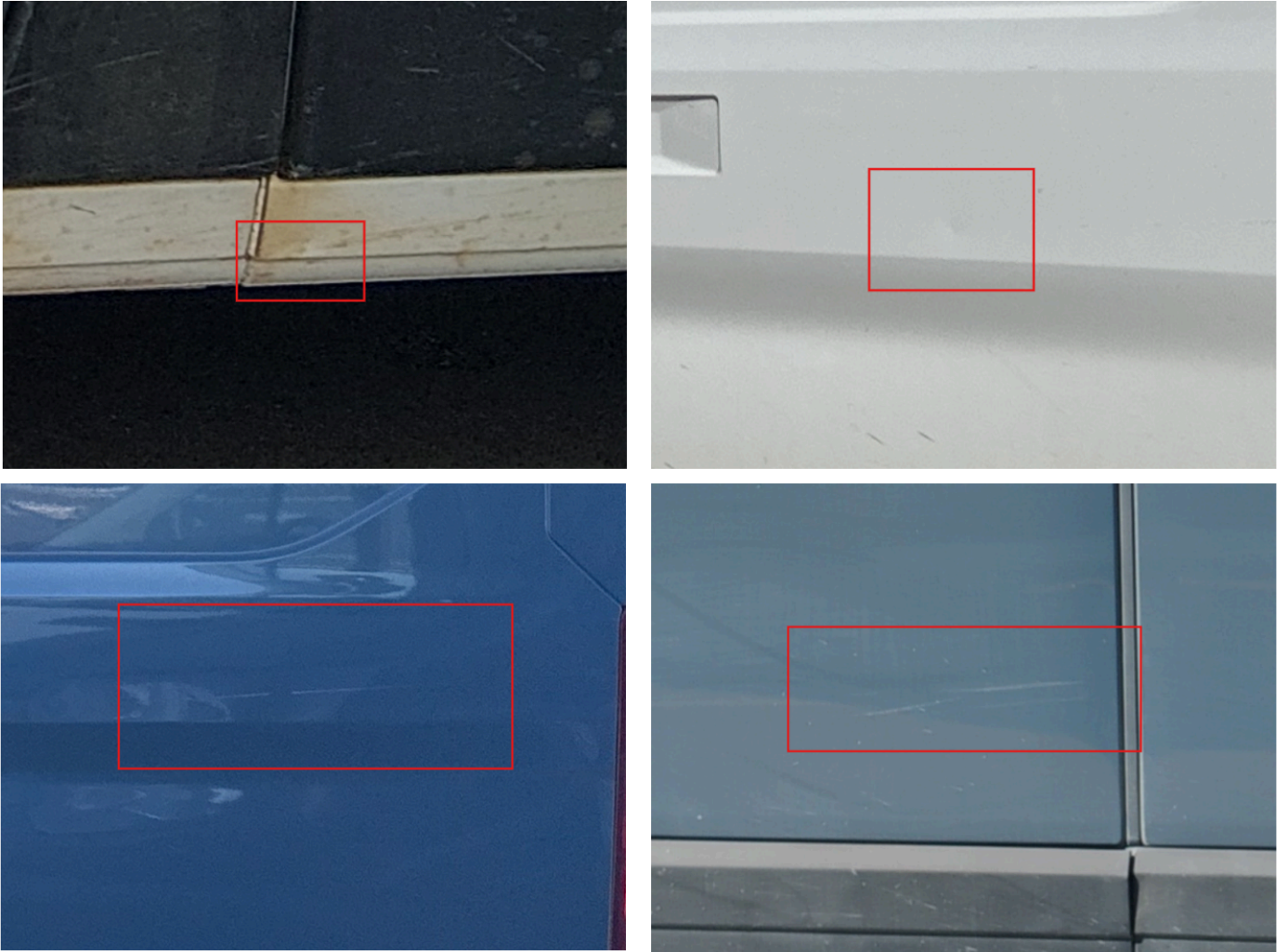
4. What is "New Damage"?

Type	Definition
Absolute New Damage	Damage that newly occurred on the current quote and was not present in any prequote.
Missed New Damage	Damage that actually occurred earlier but was missed in previous inspections . In our context, we still treat and report this as "New" .
Overlapping Damage	Even if new damage occurs directly on top of an existing damage region , we still treat and report this as "New" .

4.1 Exception Case (For US Alone):

For US Dealer quotes, if a very minor scratch or dent has been missed for more than three prequote inspections, it is to be treated as Existing Damage (not Missed New Damage).

Very Minor Scratches and Dents



(Fig 4.1 Very Minor scratches and Dents)

5. Manual Comparison (MC Page) (Some procedures here have been updated as discussed in the meeting. This section will update shortly, until then please follow the meeting procedures. If you have doubts, please reach out to **Madhava Krishnar/ Vignesh (Vicky) M/ Saravana Kumar**)

On the MC Page, you compare **detected damages against prequote images** to classify them as **New, Existing, or FP** (False Positive).

Accurately reporting new damage to the client is the primary value we provide. This is the most critical part of our validation process.

5.1 Damage Classification Rules

Situation	Classification	Action
Damage has an existing ID	Existing	Assign it with that existing ID (see Fig 5.1).
Damage is not present in the prequote	New	Assign as New in the current quote.
Damage is in the prequote but has no ID in the MC Page or Vehicle Details	Missed New	If the Prequote Marking option is available, mark as New in the oldest available prequote. If not available, mark as New in the current quote.
Damage is FP	FP	Assign as FP.
Damage is in the prequote and has an ID being repaired in Vehicle Details	Existing	Assign as Existing with the prequote image.

Sample MC Page

Manual Comparison [Vehicle details](#) [Quote details](#)

Current Inspection

frontbumper

scratch ☒ Existing damage ☐ New damage ☐ False positive



For existing damage, select the correct existing ID and set it to Existing damage in Current Inspection.

Current Condition Existing IDs

frontbumper

0eLx7tzt0 | scratch 9dYVWU44bj | scratch ucjZDyWBLc | scratch
iulokTSDNR | bumperdent rCkgGk8FB2 | bumpertear .CpLbI-D8r | scratch

Before Images |



(Fig 5.1 Sample MC Page)

5.2 Exception Case (For US Alone)

Situation	Classification	Action
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Minor scratches/dents existing for more than 3 prequotes (see Fig 4.1)	Existing (Exception)	Can be assigned as Existing.
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5.3 Issues and Actions

Situation	Action
New Damage with wrong panel name	For Fully Manual Flow or Yellow to MC Flow: Go back to VGG and change the panel name. For MC Flow: Report it in the Validation Queries Channel and reprocess as Fully Manual to change the panel name.
Existing Damage with wrong panel name	For Fully Manual Flow or Yellow to MC Flow: Go back to VGG and change the panel name. For MC Flow: Assign as FP
LP is wrong on the MC Page	For Fully Manual Flow or Yellow to MC Flow: Go back to VGG and change the LP. For MC Flow: Report it in the Validation Queries Channel and reprocess as Fully Manual to change the LP.
New damages missed to detect in the current quote (visible on MC Page)	For Fully Manual Flow or Yellow to MC Flow: Go back to VGG and mark the missed damages. For MC Flow: Report it in the Validation Queries Channel and reprocess as Fully Manual to mark the damage(s).
New damage merged with the old one during auto-comparison	Report it in the Validation Queries Channel. Repair the old id in vehicle details. And reprocess as Fully Manual to mark it as new.
Images do not show in the " Current Inspection " section of the MC Page	Report it in the Validation Queries Channel as a Current Inspection Issue (see Fig 5.3).
New damage with rotated image	Hold it and report it in the Validation Queries Channel.
Existing damage with rotated image	Assign as FP.

Current Inspection issue

Current Inspection

leftcpillar

Current Condition

leftcpillar

rV2Hclnnf | d2 z44SRaknp | d2 Before Images |

←




rightslidingdoorcladding

rightslidingdoorcladding

sxsVeVbdVq | scratch Before Images |

←

(Fig 5.3 Current Inspection Issue)

6. Final Step: Repair Status Update

At the end of the day, update the Repair Status for all vehicles validated in the **Repaired Damages Marking Page**.

Situation	Action
Damage has been physically repaired by the client	Set that ID to Repaired
Same damage has more than one ID assigned to it	Keep the most accurate ID ; mark all other duplicate IDs as Repaired .
Damage is an FP	Mark as Repaired to remove it from the active damage list.
Not sure whether to keep or skip a damage due to lack of evidence	Set status to Not sure to skip without discarding.

Repaired Damages Marking Page

The screenshot shows the 'Repaired Damages Marking Page' interface. On the left, there is a sidebar with a search bar and a list of vehicles. The main area displays the 'Repaired Damages Marking' section, which includes a list of damages and their status (Repaired, Not sure). A vehicle image is also displayed.

LP Number	Total Panels to be reviewed
7FCEHEB78PH015813	12
7FCEHEB20RND24111	6
23 - QC 97803	12
FMDXB01	4
7FCEHEB27WQ24103	5
7FCEHEB70PH009245	8
3CSURVVGKLE129477	3
Z007923	3
FPPH49	6
3CDARRVVG0MES36885	8

The 'Repaired Damages Marking' section shows a list of damages and their status. The status can be set to 'Repaired' or 'Not sure'. A vehicle image is also displayed.

(Fig 6.1 Repaired Damages Marking Page)

7. QA Plan - 8 Members (For Reviewers)

1. New Damage Review – 2 members (200 quotes/day - alternate days plan covers 3 days/week)

- Outcome: New Damage accuracy tracking, Identification of tech issues and Validator feedback
- Details: Damage level, Panel level, Flow level, Dealer level and Validator level

2. Validators' Overall Quote Review – 5 members (400 quotes/day)

- Outcome: **Validator feedback and close follow-up**, Validator accuracy tracking, Identification of tech issues and user side issues (image issues or non conventional cases) & **Existing Damage Tracking**
- Reporting Format: Validator level errors per quote (FN/FP with Major/Minor classification), Dealer level errors per quote (FN/FP with Major/Minor classification), Dealer level issues and frequency.  Rough Sheet

3. ML Benchmarking – 1 member (30 quotes/day)

- Outcome: Precision & Recall (P&R) metrics for Model performance for specific dealers, bodystyles, and environments - based on requirements

For any ad hoc tasks, we will proportionally reduce the daily validator review volume and reallocate bandwidth to the assigned tasks.

8. Manuals and others: